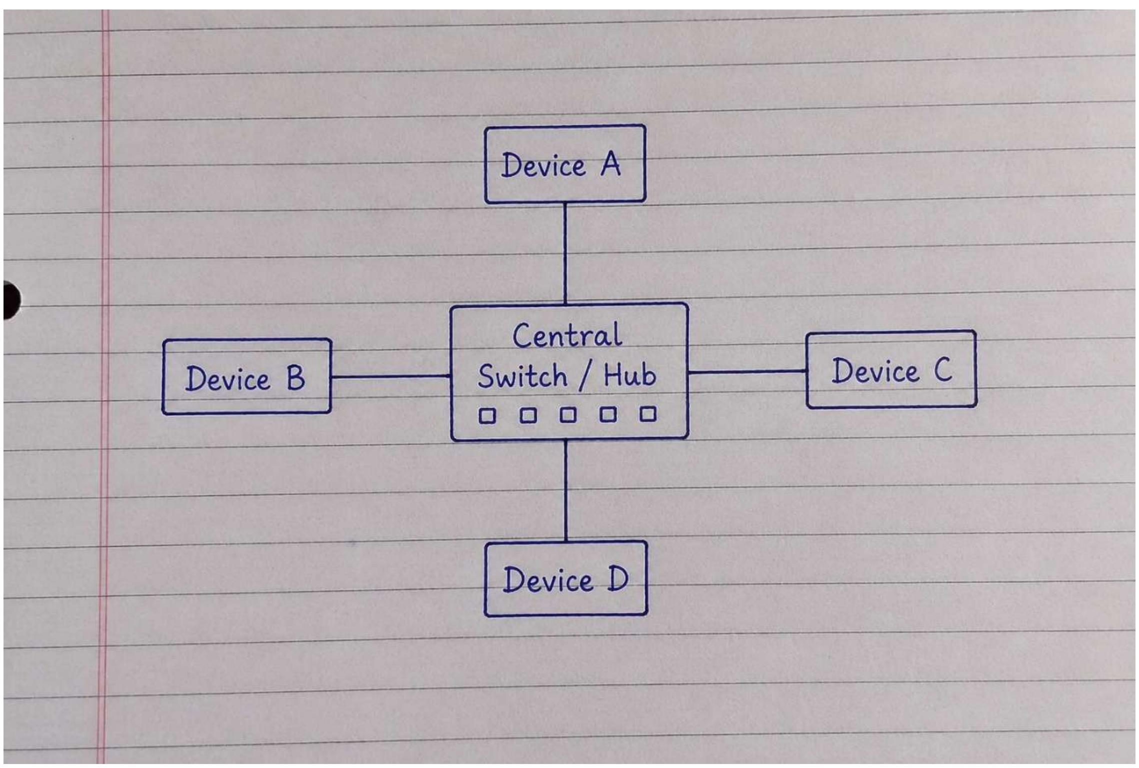
**Task 1:**

**Star, Bus, and Mesh Topologies (Diagrams & Real-World Apps)**

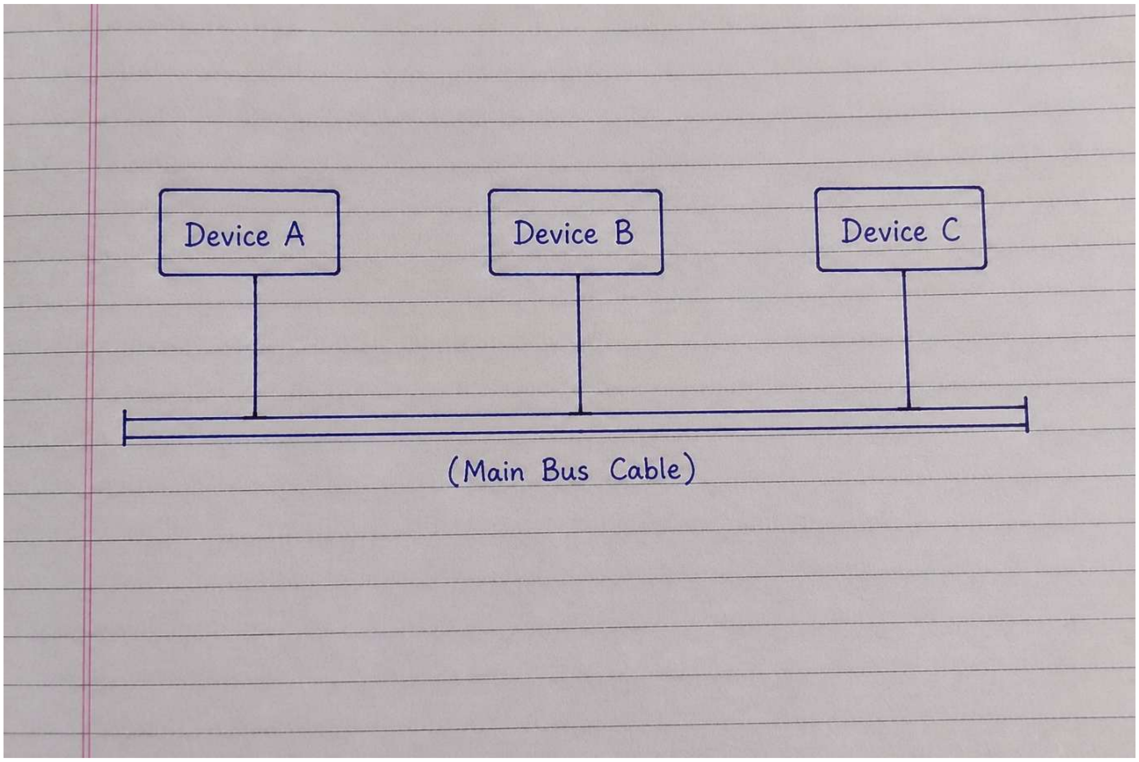
**Objective:** Understand structural designs of Star, Bus, and Mesh network topologies, draw their structures, and associate them with real-world applications.

## 1. Text-Based Network Topology Diagrams:

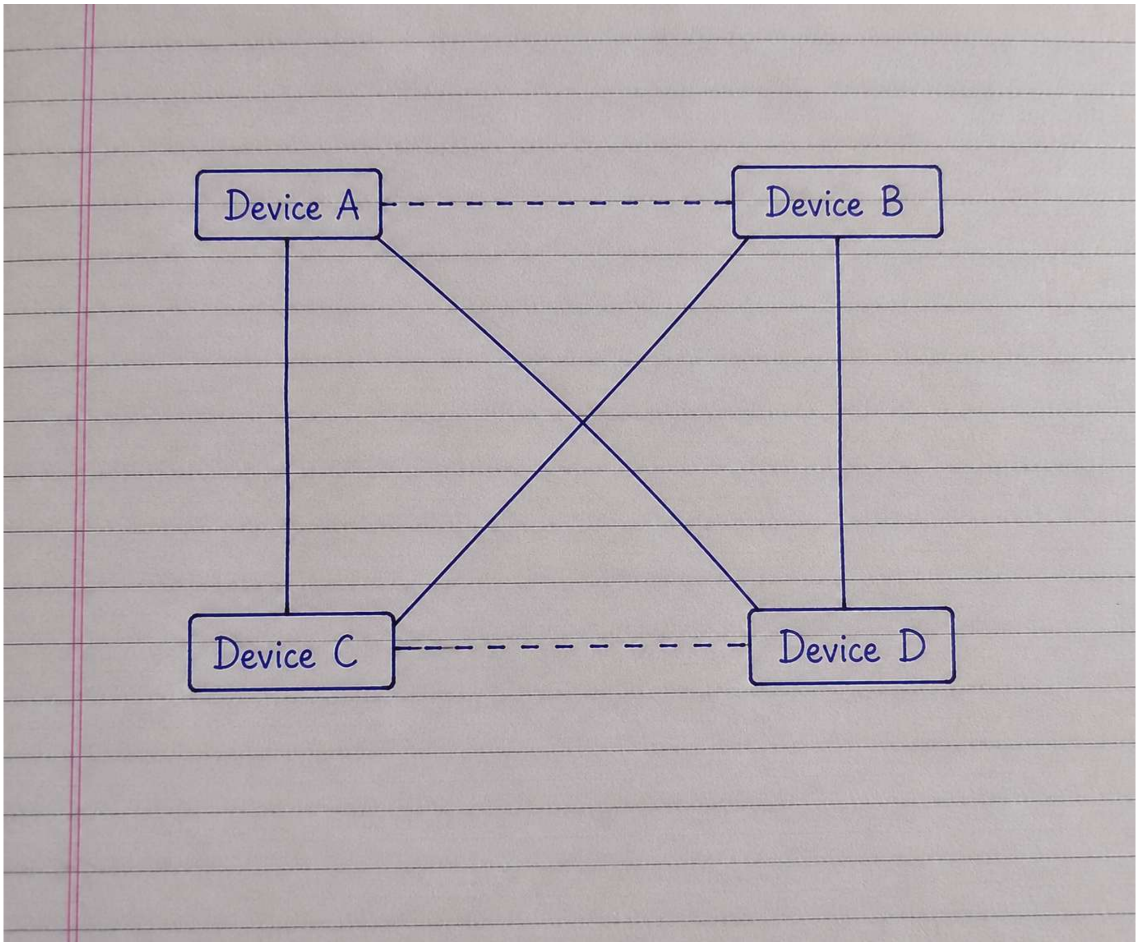
### Star Topology Diagram:



Bus Topology Diagram:



Mesh Topology Diagram:




**2. Real-World Applications & Justification:**

1. **Star Topology**

    - **Real-World App/Service: IPL Live Streaming (JioCinema / Disney+ Hotstar)**

- **Why:** All client devices connect to a central streaming server; if one client loses connection, it does not disrupt the live video feed for other connected users.

2. **Bus Topology**

   - **Real-World App/Service: Small Warehouse / Local Store Billing System (Flipkart Express Hub)**
   - **Why:** Requires minimal cabling and low setup cost to connect a few barcode scanners and desktop computers along a single backbone cable.

3. **Mesh Topology**

   - **Real-World App/Service: WhatsApp P2P Voice/Video Calling & Military Messaging**
   - **Why:** Devices connect directly to each other via multiple redundant links, ensuring zero connection dropouts and high fault tolerance if a single network path fails.

**Task 2:**

**Network Topologies Comparison Table**

**Objective:** Compare Star, Bus, Ring, and Mesh topologies by listing 4 pros, 4 cons, best use cases, and real-world app examples for each.

| Topology | Pros (At least 4) | Cons (At least 4) | Best Use Case & Real-World App Example |
|---|---|---|---|
| **Star Topology** | 1. Easy to install and manage.<br><br>2. Single device failure doesn't crash the network.<br><br>3. Easy to add new devices | 1. Central switch/hub is a single point of failure.<br><br>2. Requires more cable length than Bus. | **Best Use Case:** Client-Server applications with central control.<br><br>**App Example: Instagram / Zomato** |

| Topology | Pros (At least 4) | Cons (At least 4) | Best Use Case & Real-World App Example |
|---|---|---|---|
| | without network downtime.<br><br>4. Centralized network monitoring and management. | 3. Switch cost increases overall budget.<br><br>4. Performance depends entirely on central switch capacity. | (Central servers handling millions of user requests). |
| **Bus Topology** | 1. Very cheap and low cost to implement.<br><br>2. Uses minimal cable length. | 1. Main backbone cable failure brings down entire network.<br><br>2. Slow performance under heavy | **Best Use Case:** Small offices or low-budget networks.<br><br>**App Example: Paytm** |

| Topology | Pros (At least 4) | Cons (At least 4) | Best Use Case & Real-World App Example |
|---|---|---|---|
| | 3. Easy to set up for small/temporary networks.<br><br>4. Does not require active hardware like switches. | traffic (data collisions).<br><br>3. Difficult to identify and isolate faults.<br><br>4. Performance degrades rapidly as nodes increase. | **Standee / Local Store POS** (Basic offline/local barcode billing system). |
| **Ring Topology** | 1. Equal data access for all devices (Token passing). | 1. Break in the main cable loop disables entire network. | **Best Use Case:** Legacy high-traffic fiber loops (FDDI networks). |

| Topology | Pros (At least 4) | Cons (At least 4) | Best Use Case & Real-World App Example |
|---|---|---|---|
| | 2. Performs better than Bus under heavy traffic.<br><br>3. No data packet collisions.<br><br>4. Orderly network structure without central hub. | 2. Slow data transfer (packets pass through every intermediate device).<br><br>3. Adding/removing nodes disrupts network service.<br><br>4. Complex troubleshooting due to unidirectional flow. | **App Example: Internal Banking Transaction Queue** (Sequential step-by-step verification). |

| Topology | Pros (At least 4) | Cons (At least 4) | Best Use Case & Real-World App Example |
|---|---|---|---|
| **Mesh Topology** | 1. Extremely reliable with high fault tolerance.<br><br>2. No data traffic congestion (Dedicated point-to-point links).<br><br>3. High security and privacy.<br><br>4. Single device failure has zero | 1. Extremely expensive due to huge cable requirements.<br><br>2. Complex installation and initial configuration.<br><br>3. Requires heavy hardware (multiple NIC ports per node). | **Best Use Case:** Mission-critical and peer-to-peer networks.<br><br>**App Example: WhatsApp P2P Encrypted Video Calls / PUBG Multiplayer** (Direct zero-downtime node connections). |

| Topology | Pros (At least 4) | Cons (At least 4) | Best Use Case & Real-World App Example |
|---|---|---|---|
| | impact on network. | 4. High maintenance overhead. | |

## Task 3:

## Network Topology Selection for Multiplayer Gaming (PUBG / Free Fire)

**Objective:** Select and justify the optimal network topology for hosted multiplayer game servers, evaluating performance impacts on speed, reliability, and cost.

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### Selected Topology: Star Topology (Client-Server Architecture)

**Justification & Performance Impact:** For a high-action multiplayer game like PUBG or Free Fire, **Star Topology** is the ideal choice where all player devices (clients) connect directly to a high-speed central game server. In terms of **speed**, it delivers ultra-low latency and fast packet processing required for real-time synchronization. Regarding **reliability**, if a single player loses connection or experiences lag, it does not affect the game state for other active players. While the setup and maintenance **cost** of high-performance central servers is high, it is essential for anti-cheat management and seamless multiplayer stability.

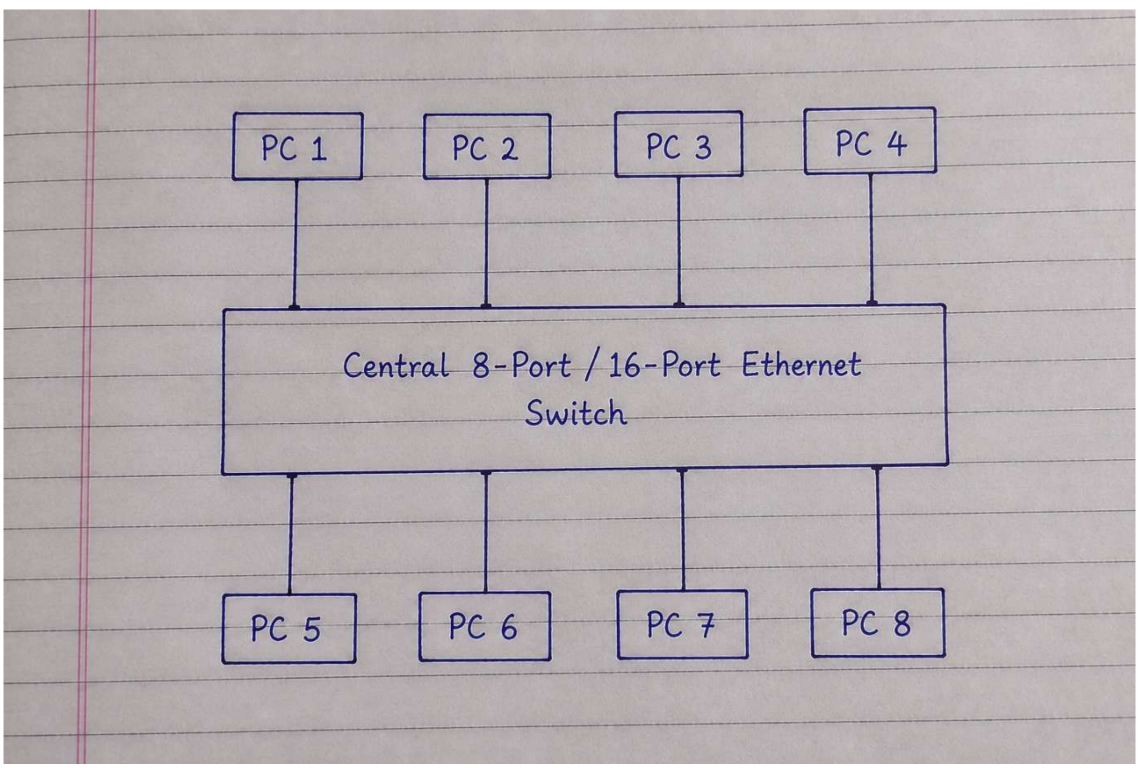
**Task 4:**

**Warehouse Network Design Scenario**

**Objective:** Design a reliable, cost-effective, and scalable network topology for connecting 8 workstations in a small e-commerce warehouse.

**1. Recommended Topology: Star Topology**

2. Quick Network Diagram:



### 3. Reasoning & Justification:

- **Cost-Effectiveness:** A central 8-port or 16-port Ethernet switch is inexpensive and uses simple Cat6 cabling, keeping initial infrastructure costs low for a small warehouse.

- **Ease of Troubleshooting & Reliability:** If any single computer or cable fails, only that specific workstation goes offline while the remaining 7 computers continue uninterrupted order processing and inventory updates.

- **Future Expansion:** Adding new computers or barcode printers in the future requires simply plugging a new cable into the central switch without disrupting active operations.